

ZKP-SPARQL: Zero-Knowledge Proofs for SPARQL Query Verification

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A Appendix

This appendix presents the theorem and proof for the Primitive BGP gadget.

Theorem 1 (Primitive BGP correctness and completeness).

Let tp be a triple pattern evaluated over an RDF graph $\mathcal{G}$. Let $\Omega_{tp}^* = \{\lambda_{tp}(t) \mid t \in \mathcal{G} \wedge t \models tp\}$ be the intended multiset of solution mappings induced by the triples of $\mathcal{G}$ matching tp . Let Λ be the table generated by the row-wise sub-circuit of the Primitive BGP gadget, and let M be the table obtained by encoding the engine output Ω_{tp} in the same variable order and with the same padding convention. If the circuit constraints are satisfied and the multiset-equality check enforces $\Lambda \equiv M$, then the engine output Ω_{tp} is correct and complete with respect to tp , i.e., $\Omega_{tp} = \Omega_{tp}^*$ as multisets.

Proof Theorem 1. Let us recall that Ω_{tp}^* denotes the intended multiset, $\Omega_{tp}^* = \{\lambda_{tp}(t) \mid t \in \mathcal{G} \wedge t \models tp\}$, where $\lambda_{tp}(t)$ is the solution mapping induced by the matching triple t . The table Λ is the circuit-internal representation of Ω_{tp}^* , whereas M is the table obtained by encoding the engine output Ω_{tp} in the same variable order and with the same padding convention.

First, we prove completeness by contradiction. Assume that Ω_{tp} is not complete, i.e., there exists a mapping μ such that $\mu \in \Omega_{tp}^*$ but $\mu \notin \Omega_{tp}$. Since $\mu \in \Omega_{tp}^*$, there exists a triple $t' \in \mathcal{G}$ such that $t' \models tp$ and $\mu = \lambda_{tp}(t')$. The row-wise part of the Primitive BGP gadget scans every triple in $\mathcal{G}$; hence, there exists a row i corresponding to t' . Since $t' \models tp$, the matching constraints force $E_s = E_p = E_o = 0$, and therefore the match indicator satisfies $m_i = 0$. By the constraints on the tuple representation of graph-induced mappings, the row Λ_i is then forced to encode $\lambda_{tp}(t') = \mu$, rather than a dummy tuple. Hence μ appears in Λ with the multiplicity induced by the matching triples in $\mathcal{G}$.

However, M encodes Ω_{tp} . Since, by assumption, μ is missing from Ω_{tp} , the multiplicity of μ in M is smaller than its multiplicity in Λ . Thus, Λ and M cannot be equal as multisets. This contradicts the multiset equality $\Lambda \equiv M$. Therefore, no mapping induced by a matching triple can be omitted from Ω_{tp} , and the output is complete.

Second, we prove correctness by contradiction. Assume that Ω_{tp} is not correct, i.e., there exists a mapping μ such that $\mu \in \Omega_{tp}$ but $\mu \notin \Omega_{tp}^*$. Since M encodes Ω_{tp} , the mapping μ appears in M . Because the circuit enforces $\Lambda \equiv M$ as multiset equality, μ must also appear in Λ with the same multiplicity. Therefore, there exists a non-dummy row i of Λ encoding μ .

By construction of Λ , a non-dummy row can be produced only when the corresponding match indicator satisfies $m_i = 0$. The matching constraints imply that $m_i = 0$ only if the triple t_i assigned to that row satisfies the triple pattern, i.e., $t_i \models tp$. Moreover, the tuple-representation constraints force Λ_i to be exactly the mapping $\lambda_{tp}(t_i)$ induced by that triple. Hence $\mu = \lambda_{tp}(t_i)$ for some

$t_i \in \mathcal{G}$ such that $t_i \models tp$. By definition of Ω_{tp}^* , this implies $\mu \in \Omega_{tp}^*$, contradicting the assumption that $\mu \notin \Omega_{tp}^*$. Therefore, no spurious mapping can appear in Ω_{tp} , and the output is correct. Consequently, Ω_{tp} is both correct and complete with respect to the intended Primitive BGP semantics, and thus $\Omega_{tp} = \Omega_{tp}^*$ as multisets, including duplicate solution mappings.

For other operators, it is possible to design similar theorems, whose proofs show that, if the circuit constraints for the operator are satisfied and the multiset-equality check enforces equality between the circuit-computed table Λ and the table M encoding the engine output, then the engine output contains all and only the mappings induced by the formal SPARQL semantics of that operator, with the same multiplicities.